

LEDGER · 2026-09-01 · POLYMER CHEMISTRY / STRUCTURAL COMPOSITES

High-Impact Polydicyclopentadiene (pDCPD) Resins via Frontal Ring-Opening Metathesis Polymerization (FROMP)

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Structural Epoxy Resin
Measured advantage	300%-400% (3-4x) improvement in ballistic energy absorption (KE50) and 50% smaller delamination area compared to structural epoxy
Time to first revenue	6 months
Gross margin at parity	60.7%
Startup CapEx	\$134,500
Payback from first sale	1.9 months
Capital productivity	17.02x

Regulatory risk

Moderate — qualification costs reported (\$85,000 estimated)

Why it failed

- Margin fails the operating-cost check

Assessment

Fails unit economics under Sub-Test D. At the incumbent selling price of \$4.77/kg and unit OPEX of \$1.86/kg, the gross margin is 61.0%, failing the strict 70% threshold required at price parity.

Also flagged

- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-01 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

Terms

Published for research and educational purposes. Nothing here is investment, legal, regulatory or engineering advice, and no commercial outcome is promised or implied. Concepts are drawn from published scientific literature and have not been commercially validated; several involve chemical, biological or regulated processes that require appropriate expertise, facilities, permits and safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.